

**MEMORIA DEL PROYECTO
MOVIECARDS**

Asignatura: Integración Continua

Alumno: Daniel Saigua Labre

Fecha: 12 Julio 2026

Proceso de Creación y Despliegue de MovieCards en Azure

Stack Técnico

- Spring Boot: 2.6.7 con Java 11
- BD: H2 (desarrollo), MySQL 8.3.0 (producción - disponible)
- Build: Maven con Java 11
- Testing: JUnit 5 + Selenium (E2E con ChromeDriver)
- QA: SonarQube (localhost:9000, credentials: admin/4daniel)
- CI/CD: GitHub Actions (build → test → qa → deploy)
- Cloud: Azure Web App (moviecards-saigua en producción)

Arquitectura del Proyecto

Capas

1. Controllers MVC (MovieController, ActorController, CardController, IndexController)
2. Services (MovieServiceImpl, ActorServiceImpl, CardServiceImpl)
3. Repositories JPA (MovieJPA, ActorJPA)
4. Entities (Movie, Actor, Card - relación M2M)

Modelos

- Movie: id, title, releaseYear, duration, country, director, genre, sinopsis
- Actor: id, name, birthDate, country
- Card: idActor, idMovie (DTO para asociación)

Pipeline CI/CD en Azure (GitHub Actions)

Archivo: .github/workflows/main.yaml

4 Etapas de Ejecución:

1. BUILD (Ubuntu)

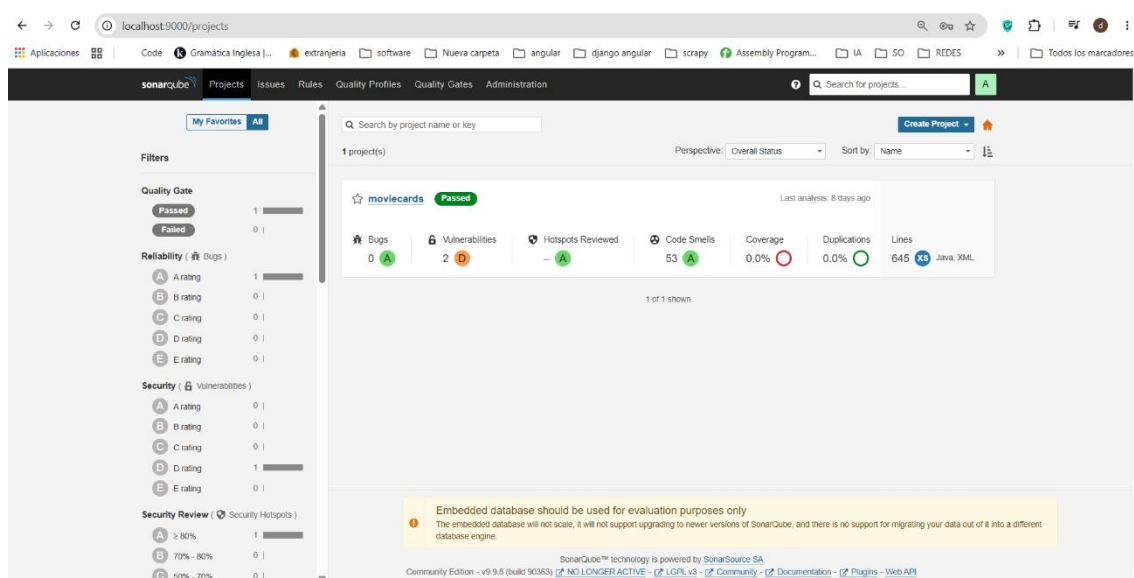
- Checkout código
- Setup JDK 11
- mvn clean package -DskipTests
- Genera artifact: moviecards-java

2. TEST (Ubuntu - Requiere BUILD)

- Instala Chrome + ChromeDriver (webdrivermanager)
- Ejecuta app: mvn spring-boot:run (espera 60s)
- Ejecuta tests: mvn clean verify
- Incluye E2E con Selenium

3. QA (Self-hosted runner - Requiere TEST)

- Análisis SonarQube: mvn sonar:sonar
- Endpoint: http://localhost:9000
- Auth: admin/4daniel
- continue-on-error: true (no bloquea deploy)



4. DEPLOY (Ubuntu - Solo rama 'master')

- Descarga artifact (JAR)
- Azure WebApp Deploy action
- App Name: moviecards-saigua
- Slot: Production
- Usa secret: AZUREAPPSERVICE_PUBLISHPROFILE_...
- Condición: if: github.ref == 'refs/heads/master'

Pasos Realizar Despliegue

1. Clonar repo: `git clone https://github.com/[user]/moviecards.git`
2. Setup local:
`mvn clean compile`
`mvn spring-boot:run`
Acceso: `http://localhost:8089`
3. Hacer commit a rama master
4. Push: `git push origin master`
5. GitHub Actions ejecuta automáticamente
6. Ver progreso en Actions tab
7. Verificar en Azure: `moviecards-saigua.azurewebsites.net`

Estructura del Flujo de Despliegue

Push master → BUILD (Compila JAR)

↓

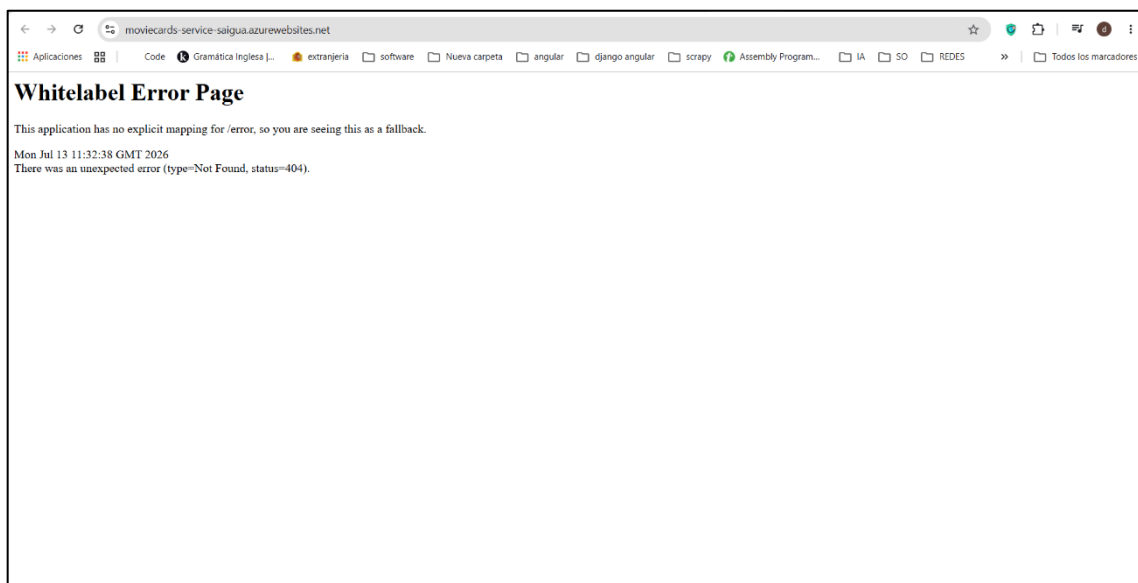
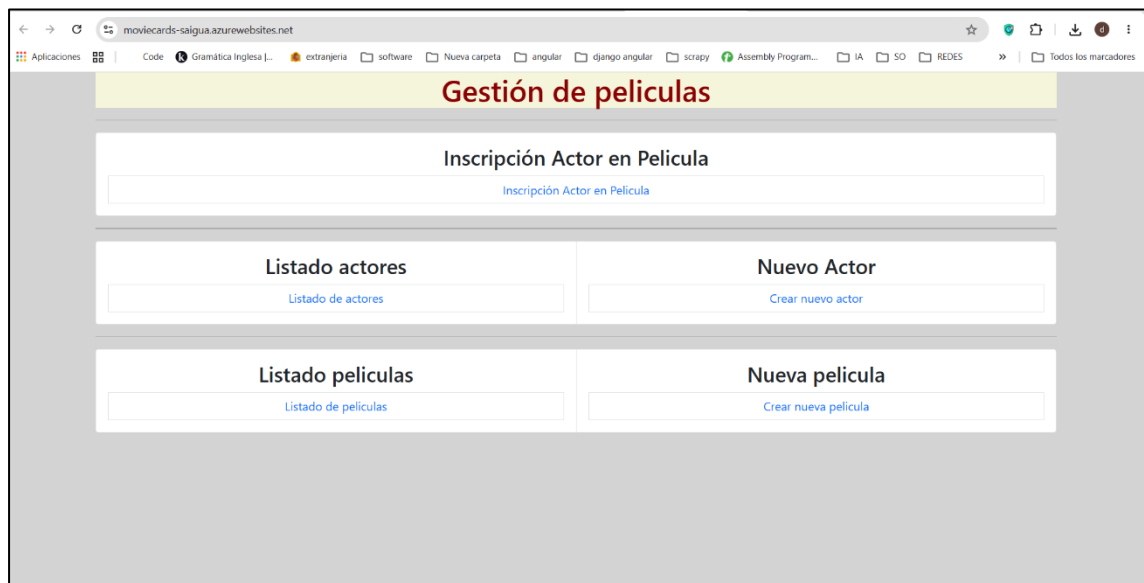
TEST (E2E con Selenium, Chrome)

↓

QA (SonarQube analysis)

↓

DEPLOY (Azure WebApp)



URLs de Referencia

- App en Azure: <https://moviecards-saigua.azurewebsites.net>
- GitHub Repo: [https://github.com/\[user\]/moviecards](https://github.com/[user]/moviecards)
- SonarQube Local: <http://localhost:9000> (admin/4daniel)
- Azure Portal: <https://portal.azure.com> → Web Apps

Testing con Postman

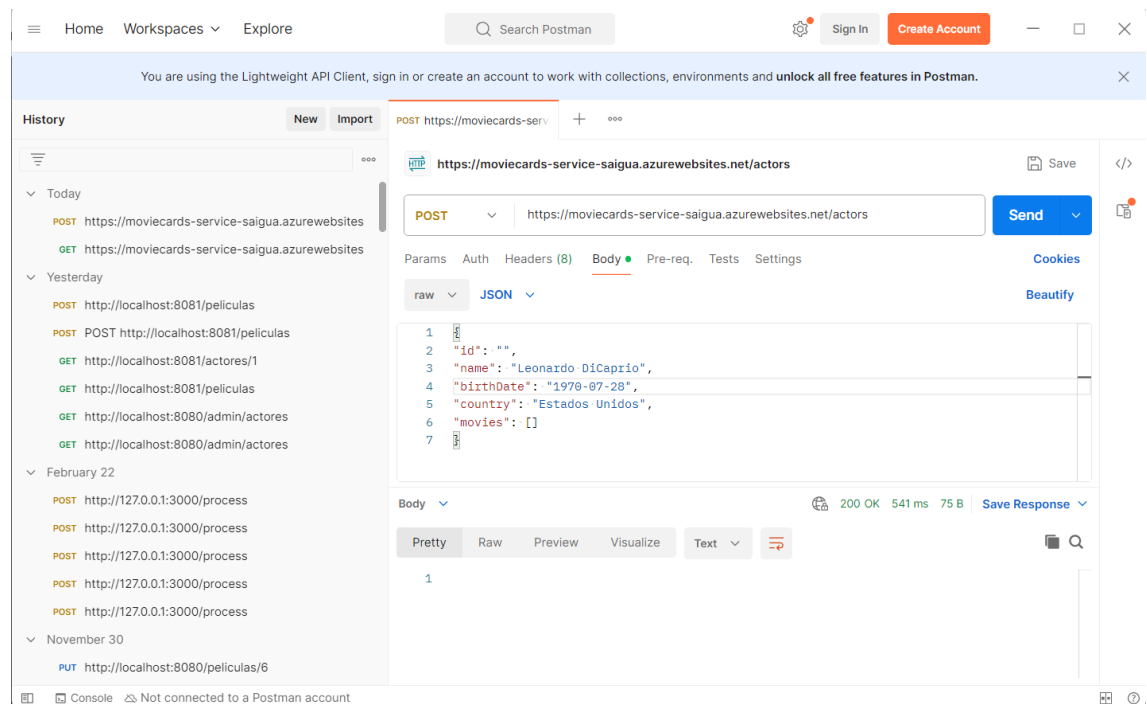
Endpoints para probar:

- GET <http://localhost:8089/movies> - Listado películas
- GET <http://localhost:8089/actors> - Listado actores
- POST <http://localhost:8089/saveMovie> - Crear película
- POST <http://localhost:8089/saveActor> - Crear actor
- POST <http://localhost:8089/register> - Asociar actor-película

Headers: Content-Type: application/json

Ejemplo JSON (película):

```
{ "title": "Avatar", "releaseYear": 2009, "duration": 162, "country": "USA", "director": "James Cameron", "genre": "Sci-Fi", "sinopsis": "En Pandora..." }
```



Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

GET https://moviecards-service-saigua.azurewebsites.net/actors/1

GET https://moviecards-service-saigua.azurewebsites.net/actors/1

Params Auth Headers (8) Body Pre-req. Tests Settings Cookies

Query Params

Key	Value	Bulk Edit
Key	Value	

Body 200 OK 456 ms 234 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "id": 1,
3   "name": "Leonardo DiCaprio",
4   "birthDate": "1970-07-28T00:00:00.000+00:00",
5   "country": "Estados Unidos",
6   "movies": []
7 }
```

Console Not connected to a Postman account

November 30

Console Not connected to a Postman account

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

POST https://moviecards-service-saigua.azurewebsites.net/movies

POST https://moviecards-service-saigua.azurewebsites.net/movies

Params Auth Headers (8) Body Pre-req. Tests Settings Cookies Beautify

raw JSON

```
1 {
2   "id": "",
3   "title": "Joker 2",
4   "releaseYear": "2024",
5   "duration": "138",
6   "country": "Estados Unidos",
7   "director": "Quentin Tarantino",
8   "genre": "Thriller",
9   "sinopsis": "Secuela de la película Joker 2019",
10  "actors": []
11 }
```

Body 200 OK 754 ms 75 B Save Response

Pretty Raw Preview Visualize Text

1

Console Not connected to a Postman account

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

GET https://moviecards-service-saigua.azurewebsites.net/movies

GET https://moviecards-service-saigua.azurewebsites.net/movies

Params Auth Headers (8) Body Pre-req. Tests Settings

raw JSON Beautify

```
2 "id": 2,
3 "title": "Joker 2",
4 "releaseYear": "2024",
5 "duration": "138",
6 "country": "Estados Unidos",
7 "director": "Quentin Tarantino",
8 "genre": "Thriller",
9 "sinopsis": "Secuela de la película Joker 2019",
10 "actors": []
```

Body 200 OK 64 ms 315 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "id": 2,
3   "title": "Joker 2",
4   "releaseYear": "2024",
5   "duration": "138",
6   "country": "Estados Unidos",
7   "director": "Quentin Tarantino",
8   "genre": "Thriller",
9   "sinopsis": "Secuela de la película Joker 2019",
10  "actors": []
11 }
```

Console Not connected to a Postman account

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import

GET https://moviecards-service-saigua.azurewebsites.net/movies

GET https://moviecards-service-saigua.azurewebsites.net/movies

Params Auth Headers (8) Body Pre-req. Tests Settings

raw JSON Beautify

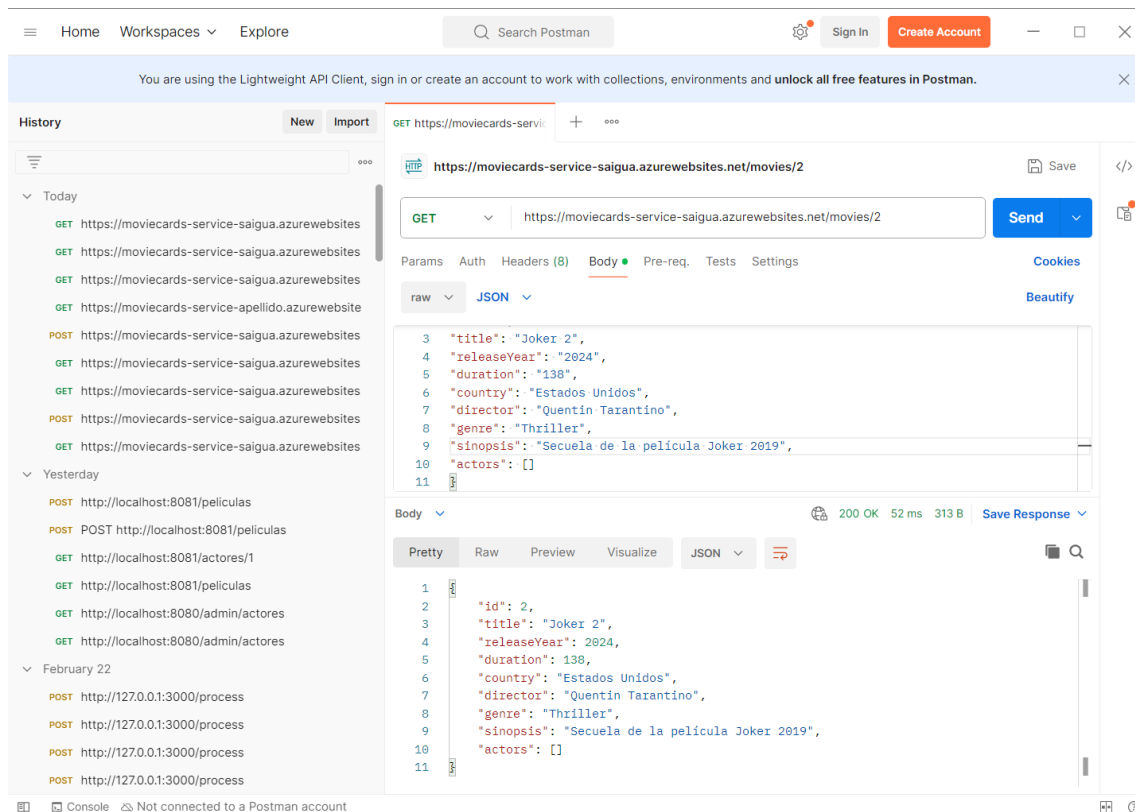
```
2 "id": 2,
3 "title": "Joker 2",
4 "releaseYear": "2024",
5 "duration": "138",
6 "country": "Estados Unidos",
7 "director": "Quentin Tarantino",
8 "genre": "Thriller",
```

Body 200 OK 44 ms 421 B Save Response

Pretty Raw Preview Visualize JSON

```
3 "id": 2,
4 "title": "Joker 2",
5 "releaseYear": "2024",
6 "duration": "138",
7 "country": "Estados Unidos",
8 "director": "Quentin Tarantino",
9 "genre": "Thriller",
10 "sinopsis": "Secuela de la película Joker 2019",
11 "actors": [
12   {
13     "id": 1,
14     "name": "Leonardo DiCaprio",
15     "birthDate": "1978-07-28T00:00:00.000+00:00",
16     "country": "Estados Unidos"
17   }
18 ]
```

Console Not connected to a Postman account



Problemas Encontrados

1. Cambio de Artifacts en GitHub Actions

Problema: El nombre del artifact generado varía según versión de pom.xml

- Inicialmente generaba: moviecards-0.0.1-SNAPSHOT.jar
- En el workflow se referenciaba erróneamente como: moviecards-java

Solución: Actualizar a actions/upload-artifact@v4 y actions/download-artifact@v4

- BUILD: upload-artifact@v4 con name: moviecards-java, path: target/*.jar
- DEPLOY: download-artifact@v4 con name: moviecards-java, package: "*.jar"
- Esto asegura capturar cualquier versión del JAR generado.